Student Attendance Tracker using C

1. Introduction:

The *Student Attendance Tracker* is a console-based application developed in C that helps manage and track student attendance efficiently. It uses data structures like **arrays and structures** to store and manipulate student records.

The system allows the user to:

- Add student details
- Mark attendance
- View attendance reports
- Search student records

2. Objectives:

- To understand practical use of **Data Structures in C**
- To implement **real-world record management**
- To improve logic building using **arrays & structures**
- To design a user-friendly system

3. Data Structures Used:

- **Structure (struct)** → to store student details
- **Array of structures** → to handle multiple students
-

4. Features:

✓ Add new student
✓ Display all students
✓ Mark attendance
✓ View attendance percentage
✓ Search student by roll number
✓ Simple menu-driven UI

5. Complete C Code:

```
#include <stdio.h>
#include <string.h>
#define MAX 100
struct Student {
   int roll;
   char name[50];
   int totalClasses;
   int attended;
};
struct Student s[MAX];
int count = 0;
// Add Student
void addStudent() {
   printf("\nEnter Roll Number: ");
   scanf("%d", &s[count].roll);
   printf("Enter Name: ");
   scanf(" %[^\n]", s[count].name);
   s[count].totalClasses = 0;
   s[count].attended = 0;
```

```
count++;
    printf("✅ Student Added Successfully!\n");
}
// Display Students
void displayStudents() {
    printf("\n--- Student List ---\n");
    for(int i = 0; i < count; i++) {
        printf("Roll: %d | Name: %s\n", s[i].roll, s[i].name);
    }
}
// Mark Attendance
void markAttendance() {
    int roll, found = 0;
    printf("\nEnter Roll Number: ");
    scanf("%d", &roll);
    for(int i = 0; i < count; i++) {
        if(s[i].roll == roll) {
            found = 1;
            s[i].totalClasses++;
            int present;
            printf("Present? (1 = Yes, 0 = No): ");
            scanf("%d", &present);
            if(present == 1)
                s[i].attended++;
            printf("✅ Attendance Marked!\n");
        }
    }
    if(!found)
        printf("❌ Student Not Found!\n");
```

```
}
// View Attendance
void viewAttendance() {
    printf("\n--- Attendance Report ---\n");
    for(int i = 0; i < count; i++) {
        float percentage = 0;
        if(s[i].totalClasses > 0)
            percentage = (float)s[i].attended / s[i].totalClasses * 100;
        printf("Roll: %d | Name: %s | Attendance: %.2f%%\n",
               s[i].roll, s[i].name, percentage);
    }
}
// Search Student
void searchStudent() {
    int roll, found = 0;
    printf("\nEnter Roll Number: ");
    scanf("%d", &roll);
    for(int i = 0; i < count; i++) {
        if(s[i].roll == roll) {
            found = 1;
            printf("Found: %s\n", s[i].name);
        }
    }
    if(!found)
        printf("✗ Student Not Found!\n");
}
int main() {
    int choice;
    while(1) {
```

```
        printf("\n\n===== Student Attendance Tracker =====\n");
        printf("1. Add Student\n");
        printf("2. Display Students\n");
        printf("3. Mark Attendance\n");
        printf("4. View Attendance\n");
        printf("5. Search Student\n");
        printf("6. Exit\n");
        printf("Enter Choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1: addStudent(); break;
            case 2: displayStudents(); break;
            case 3: markAttendance(); break;
            case 4: viewAttendance(); break;
            case 5: searchStudent(); break;
            case 6: return 0;
            default: printf("Invalid Choice!\n");
        }
    }
}
```

8. Output Screens:

Take screenshots of:

- **Main menu**

```
===== Student Attendance Tracker =====
1. Add Student
2. Display Students
3. Mark Attendance
4. View Attendance
5. Search Student
6. Exit
Enter Choice:
```

- **Adding student**

```
===== Student Attendance Tracker =====
1. Add Student
2. Display Students
3. Mark Attendance
4. View Attendance
5. Search Student
6. Exit
Enter Choice: 01

Enter Roll Number: 001
Enter Name: MOHIT
✅ Student Added Successfully!
```

- **Attendance marking**

```
===== Student Attendance Tracker =====
1. Add Student
2. Display Students
3. Mark Attendance
4. View Attendance
5. Search Student
6. Exit
Enter Choice: 03

Enter Roll Number: 001
Present? (1 = Yes, 0 = No): 1
✅ Attendance Marked!
```

- **Report display**

```
===== Student Attendance Tracker =====
1. Add Student
2. Display Students
3. Mark Attendance
4. View Attendance
5. Search Student
6. Exit
Enter Choice: 04

--- Attendance Report ---
Roll: 1 | Name: MOHIT | Attendance: 100.00%
```

9. Advantages:

- Easy to use
- Saves time
- Organized data
- Scalable

10. Future Enhancements:

- GUI using C++ / Java
- Database integration
- Mobile app version